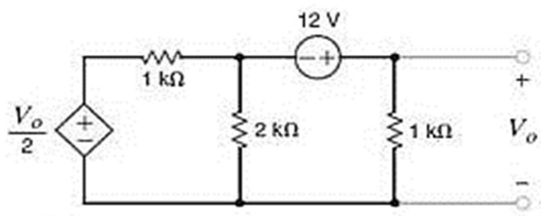
BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(END SEMESTER EXAMINATION)

CLASS: B.Tech **SEMESTER : II**
BRANCH: CSE/AIML/ECE/EEE **SESSION : SP/2024**

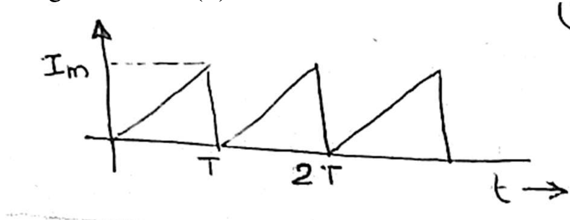
SUBJECT: EE101 BASCIS OF ELECTRICAL ENGINEERING
TIME: 3 Hours **FULL MARKS: 50**

INSTRUCTIONS:
1. The question paper contains 5 questions each of 10 marks and total 50 marks.
2. Attempt all questions.
3. The missing data, if any, may be assumed suitably.
4. Before attempting the question paper, be sure that you have got the correct question paper.
5. Tables/Data hand book/Graph paper etc. to be supplied to the candidates in the examination hall.

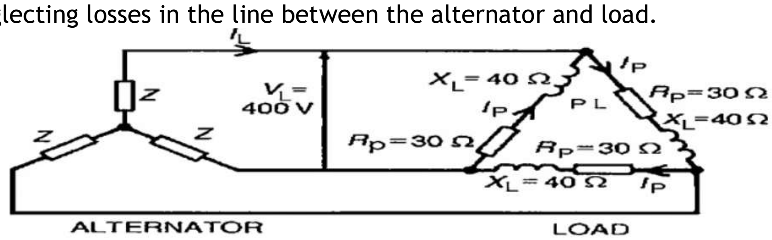
			CO	BL
Q.1(a)	Define (i) Dependent source (ii) linear element (iii) reluctance (iv) mesh (v) inductance.	[5]	1	2
Q.1(b)	For the circuit shown below, use mesh analysis to find the voltage $\mathbf{V_0}$	[5]	1	3



Q.2(a)	Determine (i) the average value and (ii) r.m.s. value of the current waveform in given below	[5]	2	3

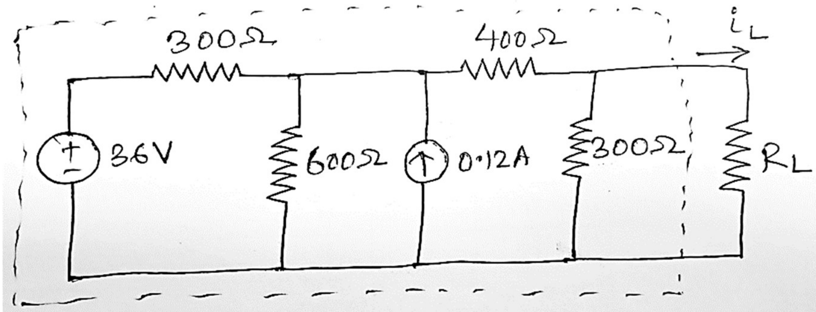


Q.2(b)	Two elements based series circuit consumes 700 W and has power factor of 0.707 (leading). If the applied voltage is $v(t)= 141\sin(314t+ 30^0)$, obtain the circuit constants. And also draw the phasor diagram.	[5]	2	4
Q.3(a)	Derive the expression for the Instantaneous Power in a balanced three phase network star or delta?	[5]	3	3
Q.3(b)	A 400V, 3-phase star connected alternator supplies a delta-connected load, each phase of which has a resistance of 30 ohms and inductive reactance 40 ohms. Calculate (a) the current supplied by the alternator and (b) the output power and the kVA of the alternator, neglecting losses in the line between the alternator and load.	[5]	3	3



PTO

Q.4(a)	Consider the following circuit shown below. Compute R_L for maximum power transfer and compute the maximum power P_{Lmax}?	[5]	4	3



Q.4(b)	A steel ring has a mean diameter of 20cm, a cross section of 25cm^2 and a radial air gap of 0.5mm cut in it. The ring is uniformly wound with 2000 turns of insulated wire and a magnetizing current of 1A produces a flux of 1.25 milli weber in the air gap. Neglecting the effect of magnetic leakage and fringing, calculate: (i) the reluctance of the magnetic circuit and (b) the relative permeability of the steel.	[5]	4	3
Q.5(a)	Explain the working principle (theory of operation) of an ideal transformer at no load.	[5]	5	4
Q.5(b)	Describe the construction of Induction type wattmeter with a suitable diagram and also draw a nice phasor diagram also.	[5]	5	2

:::::26/04/2024 E:::::